Adam Shebani

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EDUCATION

University of Wisconsin-Madison, College of Engineering — **Madison, WI**
Degree: B.S. Computer Engineering - Cumulative GPA: 3.9 — September 2025 – December 2027

Madison College — **Madison, WI**
Degree: Pre-Engineering - Cumulative GPA: 3.8 — August 2023 – May 2025

ENGINEERING EXPERIENCE

Silicon Badgers – RTL Team Designer — September 2026 – Present

- Profiled Qwen3.5-2B inference in llama.cpp (C/C++), breaking down the compute graph by operation to show matrix multiplication accounts for 90% of runtime.
- Designed a parameterizable INT4/INT8 matrix-multiplication accelerator in SystemVerilog targeting the model's attention and MLP layers.

CS 544 – Big Data Systems (Python) — UW-Madison

- Built distributed storage and analytics pipelines using Docker, HDFS, Spark, Cassandra, Kafka, and gRPC.
- Developed data processing workflows in PySpark and Parquet, and designed relational schemas and RESTful backend services.

ECE 353 – Embedded Systems and Microcontrollers (C) — UW-Madison

- Developed embedded C firmware on an ARM Cortex-M (PSoC 6) using interrupt-driven design with timers, GPIO, PWM, and ADC peripherals.
- Built concurrent I/O functionality using FreeRTOS to coordinate sampling, communication, and display updates without polling.

PROJECTS

Trading Dashboard (Python, Interactive Brokers API)

- Built a read-only market-data layer on the Interactive Brokers TWS API, wrapping its callback-driven threading in blocking calls with typed errors and tagging each quote as live, delayed, or frozen to surface data-entitlement gaps.
- Designed a local SQLite cache for historical prices that fetches only missing date ranges, validating and storing ~5,000 daily bars so repeat queries need no network access.

ML Cookbook (Python, scikit-learn, pandas)

- Built a Python library for classical ML on tabular data that infers each column's statistical type, validates analysis preconditions before running, and supports user type overrides.
- Implemented correlation, linear/logistic regression, decision tree, and random forest workflows returning immutable, JSON-serializable results, decoupling the core from notebook, desktop, or REST frontends.

Wearable Motion Tracker (Infineon PSoC 6, C, FreeRTOS)

- Brought up six peripherals over SPI, I2C, and PDM under FreeRTOS on an ARM Cortex-M4: IMU, magnetometer, pressure sensor, 60 GHz radar, MEMS microphone, and QSPI flash.
- Logged and streamed sensor data to train a CNN gesture classifier, deployed on-device at 84% accuracy.

SKILLS

- C, C++, Python, Java, SystemVerilog
- SQL, SQLite, scikit-learn, pandas
- Machine learning, Data Processing
- Claude, Codex, Git, Linux, Windows
- HDFS, Spark, Cassandra, Kafka, Docker
- AI, LLMs, Llama.cpp